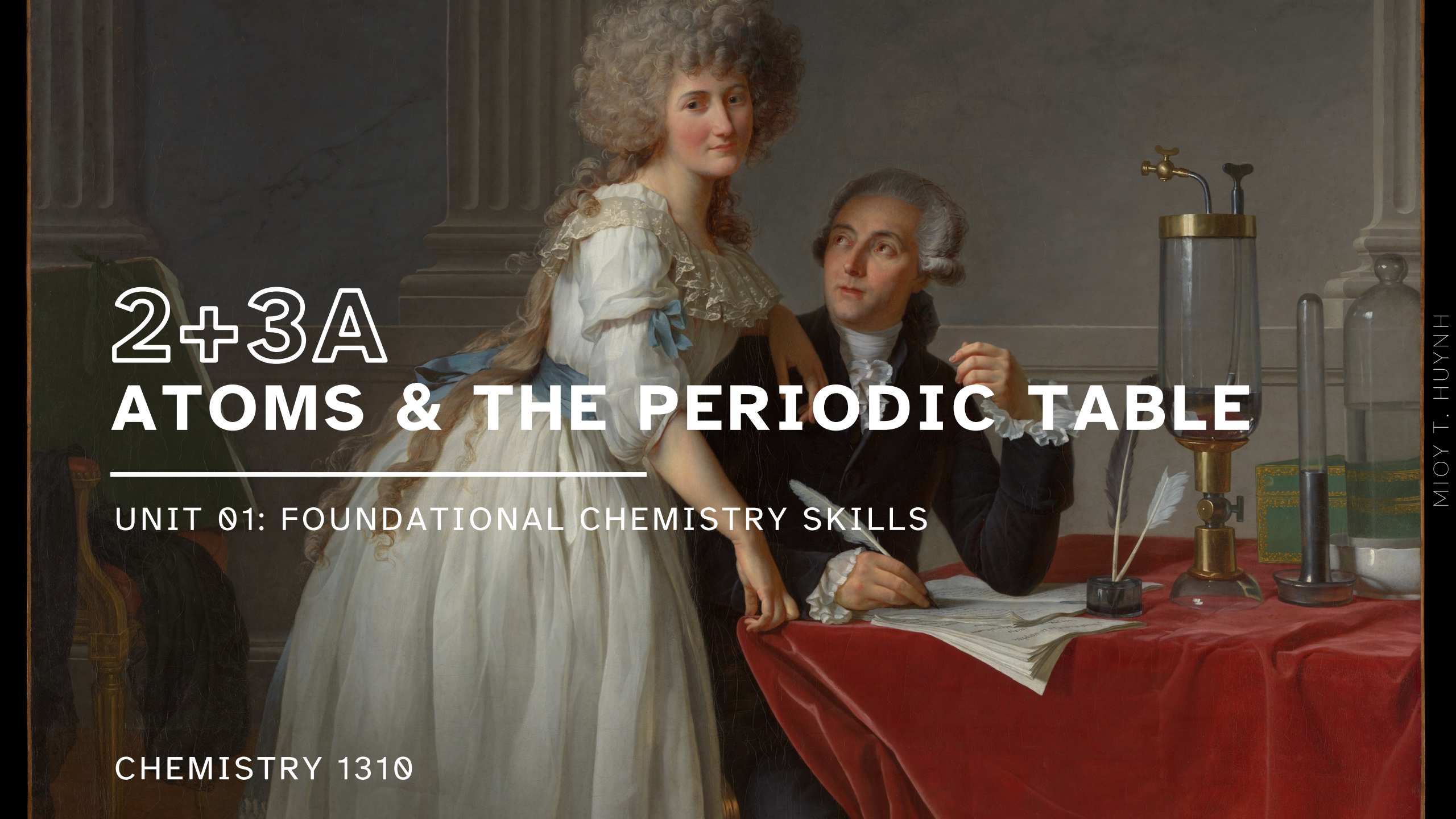


A classical painting of Antoine Lavoisier and Laplace in their laboratory. Lavoisier is seated at a table covered with a red cloth, writing on a document with a quill. Laplace stands beside him, leaning over his shoulder. The table is set with various scientific instruments, including a large glass apparatus with a pump, a thermometer, and a balance scale. The background features classical columns and a dark, atmospheric setting.

2+3A ATOMS & THE PERIODIC TABLE

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 2

1. Recognize the symbols and names of common elements.
2. Outline Dalton's atomic theory.
3. Apply Rutherford's nuclear model of the atom.
4. Identify the mass number and atomic number of an atom using the count of subatomic particles present and vice versa.
5. Identify isotopes of element using their atomic and mass numbers; count subatomic particles in a given isotope.
6. Calculate the atomic mass of an element from the masses and abundances of its naturally occurring isotopes.
7. Classify elements in a systematic manner based on their location on the periodic table.
8. Identify periods, groups, and sections of the periodic table.

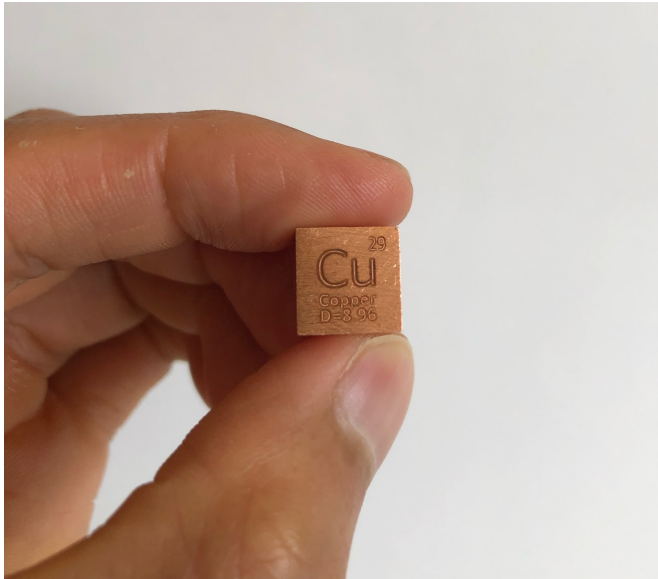
Domains of Chemistry

In this chapter, we'll discuss atomic structure, but it's important to remember that we can't actually "see" atoms.

MACROSCOPIC

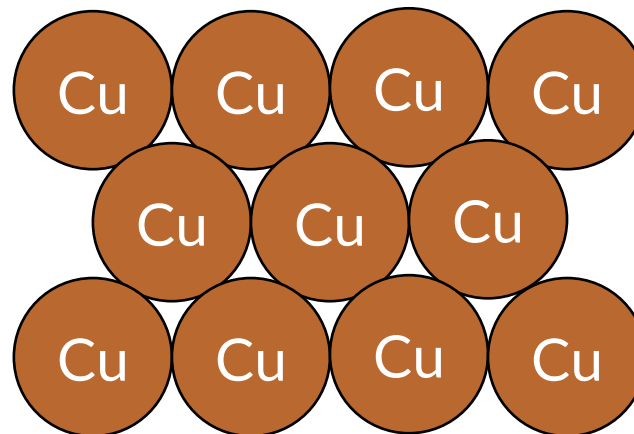
"large"

Can see with naked eye



SYMBOLIC

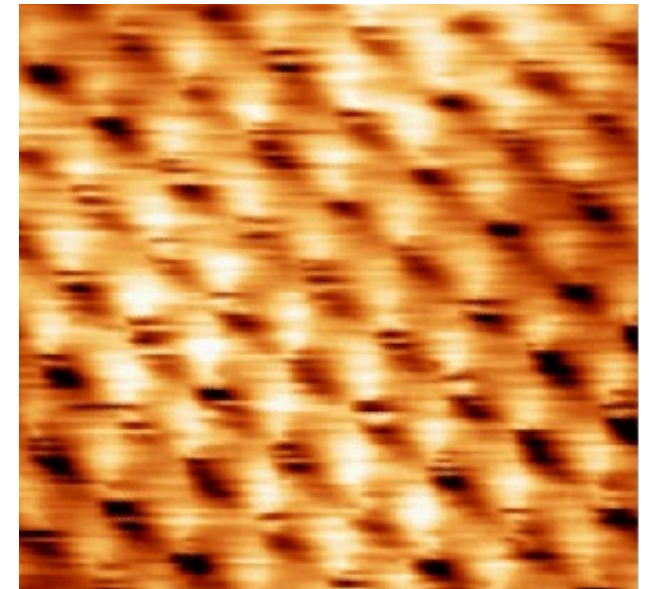
Used to portray macroscopic and microscopic domains



MICROSCOPIC

"small"

Requires equipment to "see"



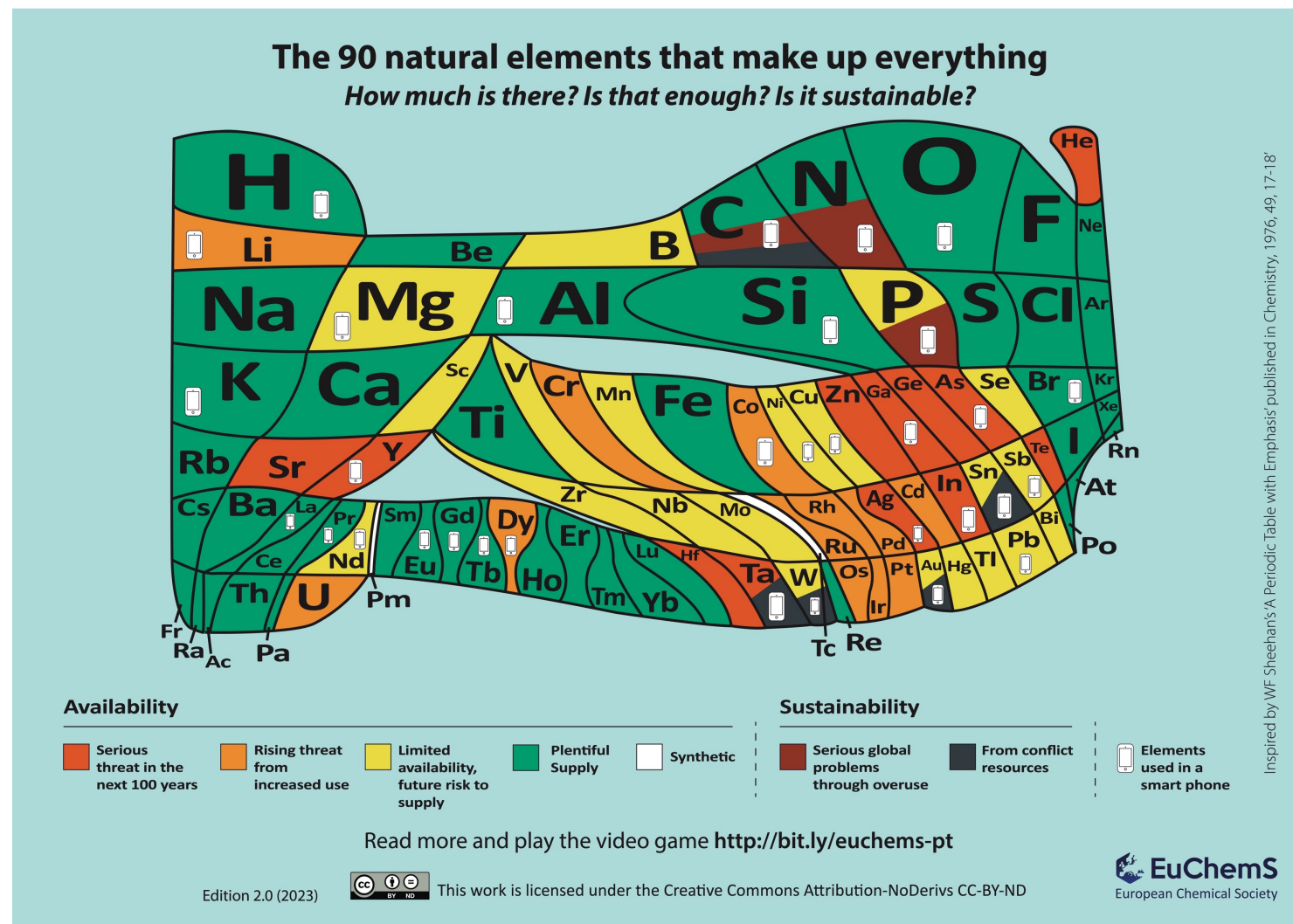
Chemical Symbols

2.1

Each of the 118 known chemical elements is identified by an internationally recognized symbol consisting of either:

- one capital letter (C, H, O, F, etc.)
- one capital letter followed by a lower-case letter (Na, Fe, Cl, Ar, etc.)

There are 10 elements whose symbols are derived from the chemical name in a language other than English.



Laws of Chemical Composition

2.2

The mass of the products generated during a chemical reaction is equal to the mass of reactants used up during the reaction.

This observation is the basis of the law of conservation of mass.

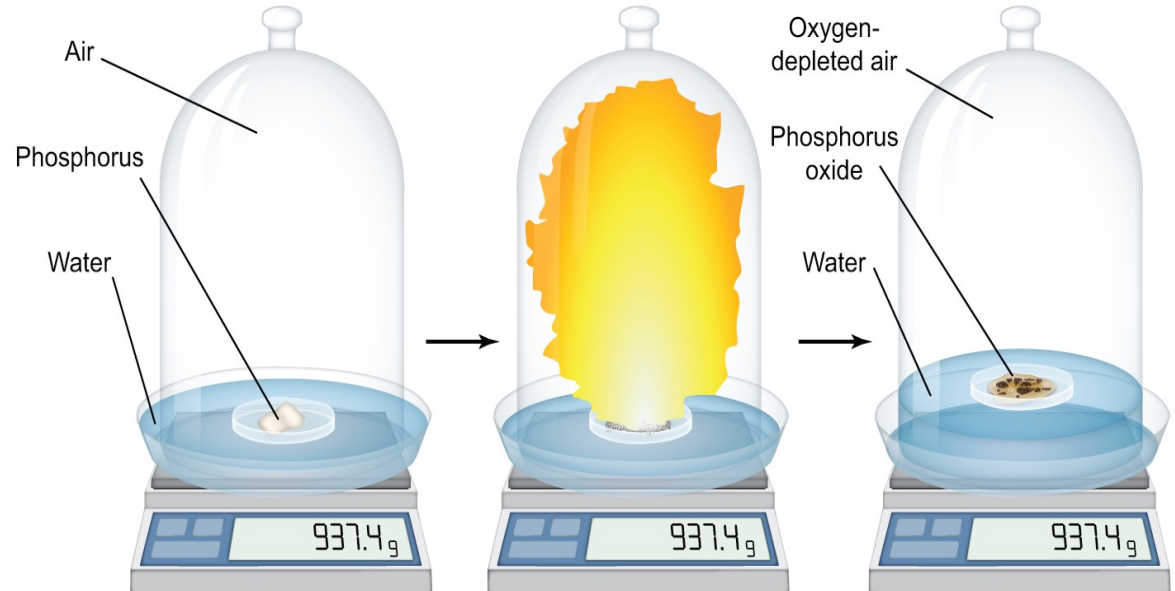


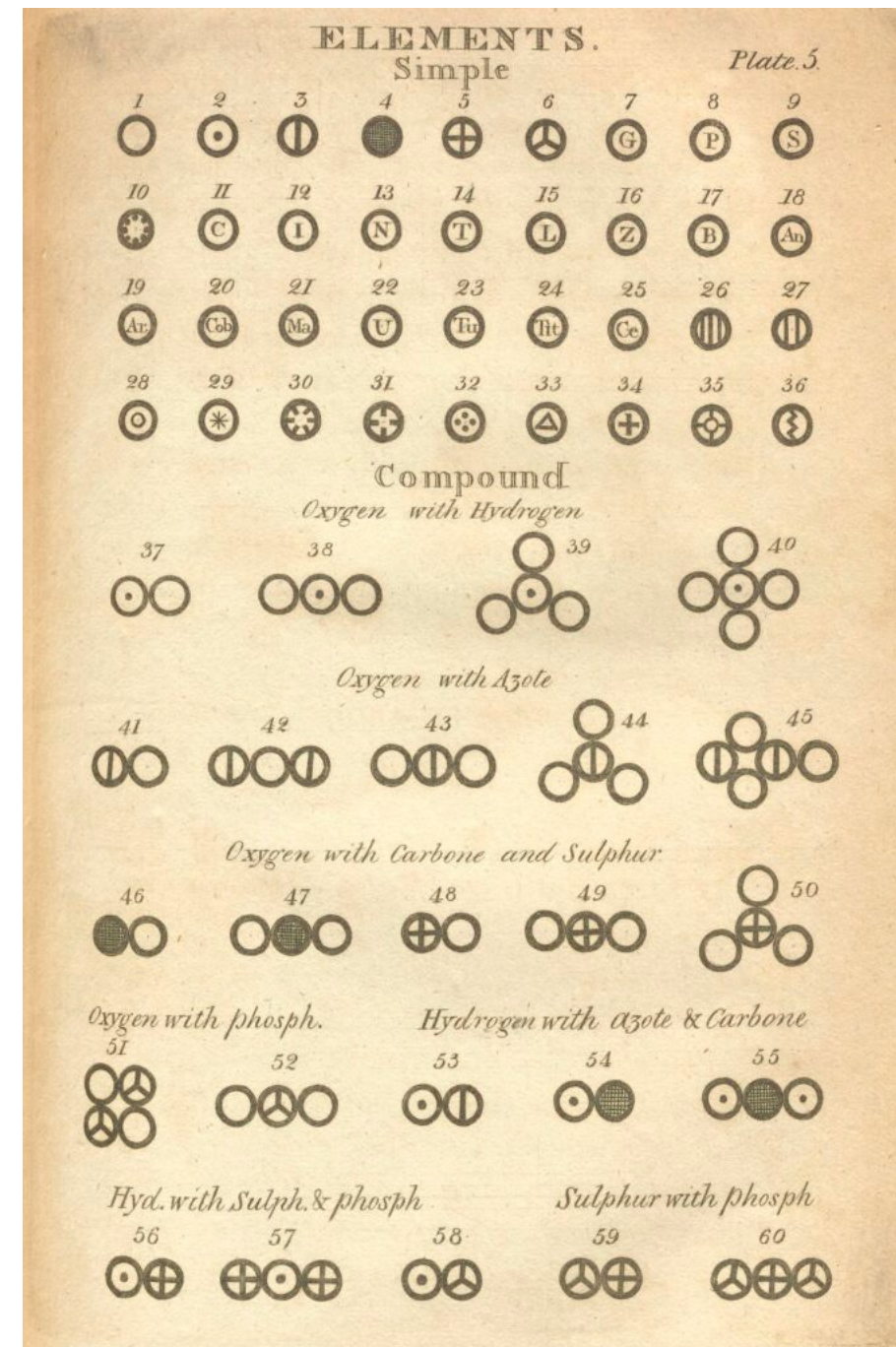
FIGURE 2.1 An Experiment to Illustrate the Law of Conservation of Mass

The History of the Atom

2.3

John Dalton formulated his atomic theory to explain the different laws of chemical combination:

1. Matter is made up of very tiny, indivisible particles called atoms.
2. Each atom of a particular element has the same mass, but the mass of an atom of one element is different from the mass of an atom of any other element.
3. Atoms combine to form what we now call molecules. When they do so, they combine in small, whole-number ratios.
4. Atoms of some pairs of elements can combine with each other in different, small, whole-number ratios to form different compounds.



The History of the Atom

2.3

English physicist J.J. Thomson discovered the negatively charged subatomic particle now known as the electron in the late 1800s.

American physicist Robert Millikan succeeded in measuring the mass of a single electron in the early 1900s.

- Because atoms are neutral overall, the “non-electron” part of the atom must have a positive charge equal to the negative charge of electrons.
- The electron accounted for only $1/1836$ of the mass of a hydrogen atom.
- These observations led to Thomson’s plum-pudding model of the atom.

Dalton’s early model (a) consisted of an indivisible sphere.

Thomson’s plum-pudding model (b) consisted of negatively charged electrons suspended in a larger sphere of positive charge.

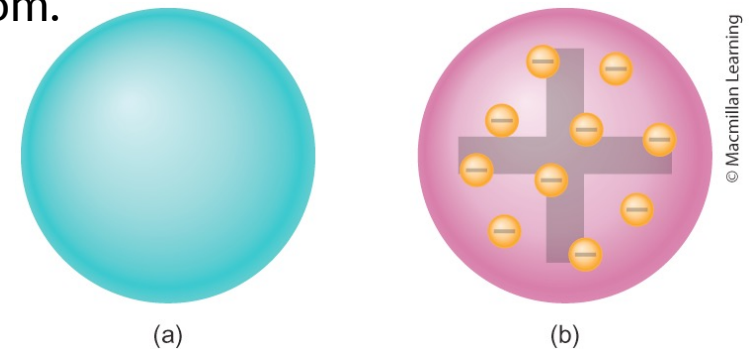


FIGURE 2.7 Dalton and Thomson’s Early Models of the Atom

The History of the Atom

2.3

British scientist Ernest Rutherford's gold foil experiment was designed to substantiate Thomson's plum-pudding model.

Instead, it led to a new nuclear model of the atom.

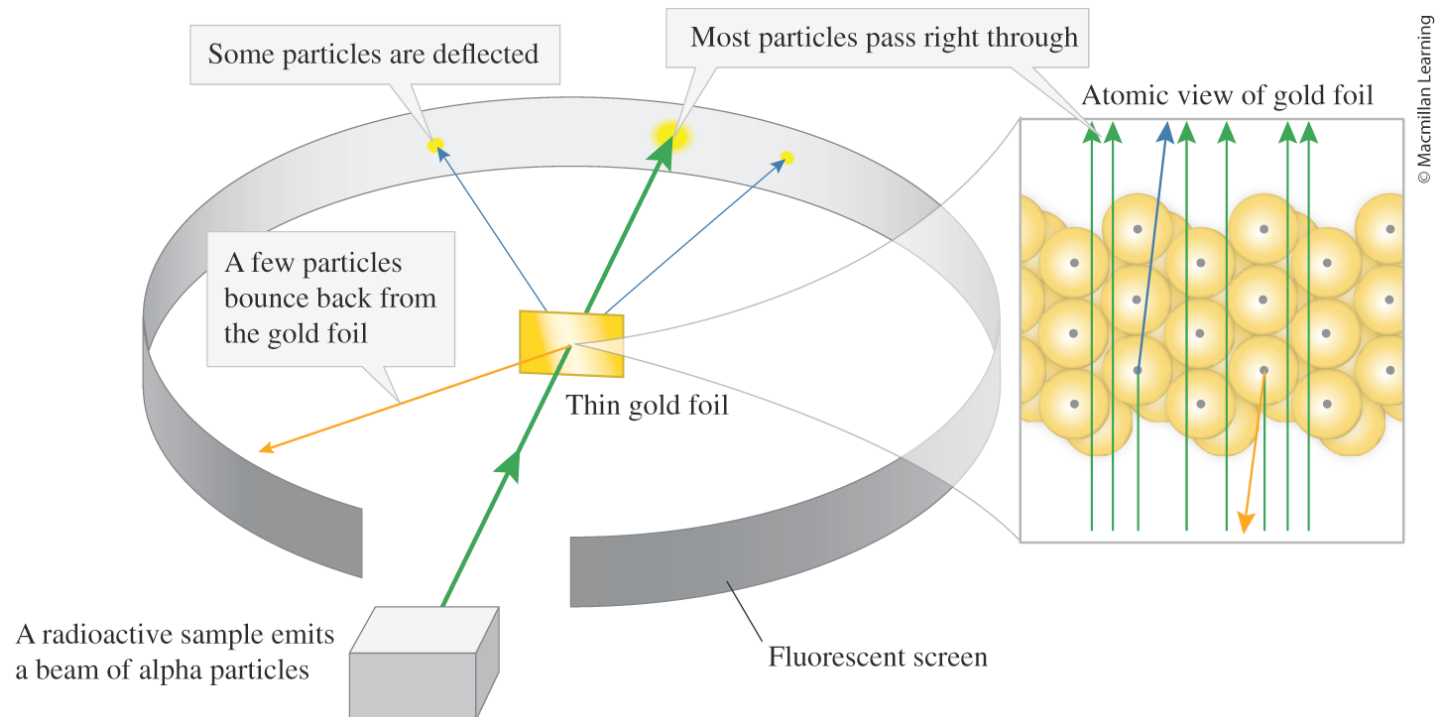


FIGURE 2.8 Rutherford's Gold Foil Experiment

The History of the Atom

2.3

Rutherford's nuclear model:

- The atom's positive charge and the majority of its mass are located in a relatively small area, the nucleus.
- The vast majority of the atom is empty space.
- Small, negatively charged electrons are spread throughout this empty space.
- The number of negatively charged electrons is equal to the number of positively charged particles (protons) inside the nucleus.

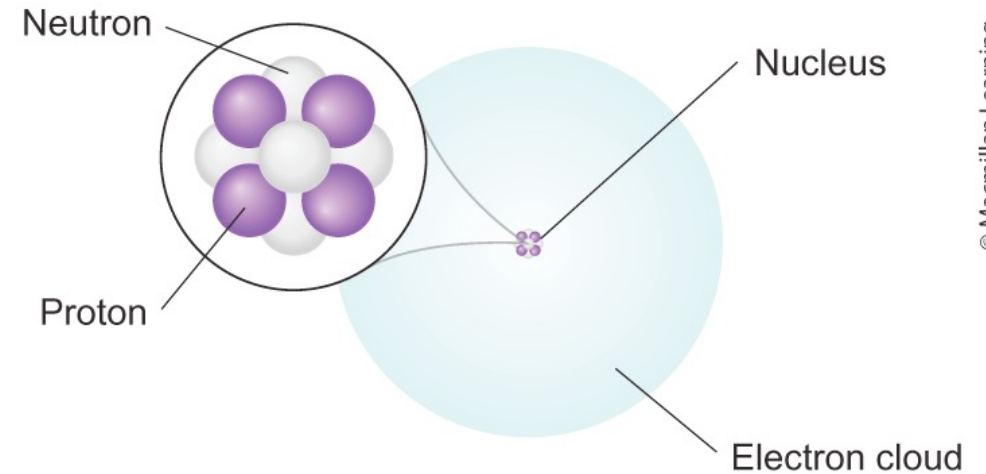


FIGURE 2.9 Nuclear Model of the Atom

Subatomic Particles, Isotopes, and Ions

2.4

Protons, neutrons, and electrons are subatomic particles.

- The nucleus comprises almost all of the *mass* of an atom whereas most of the *volume* is occupied by electrons.

Particle	Charge (e)	Mass (atomic mass units)	Location in the atom
Proton (p ⁺)	+1	1.0073	In the nucleus
Neutron (n)	0	1.0087	In the nucleus
Electron (e ⁻)	-1	0.000549	Outside the nucleus

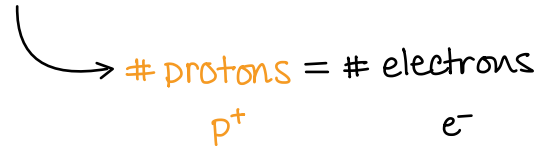
- Note: The charges given are *relative* charges, based on the charge of the electron (denoted as *e*) as the fundamental unit of charge.

$$1\ e = -1.60 \times 10^{-19}\ \text{Coulombs (C)}$$

Subatomic Particles, Isotopes, and Ions

2.4

Atoms are electrically neutral:



A curved arrow points from the text "Atoms are electrically neutral:" to the equation $\# \text{ protons} = \# \text{ electrons}$. Below "protons" is a handwritten p^+ and below "electrons" is a handwritten e^- .

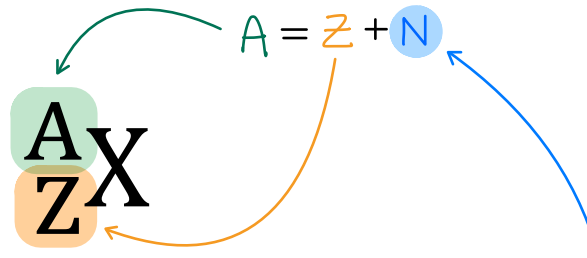
$$\# \text{ protons} = \# \text{ electrons}$$

$p^+ \quad e^-$

Atomic number, Z , is the number of protons in the nucleus of an atom.

- All atoms of a given element have the same number of protons.

i.e. Z defines element identity X



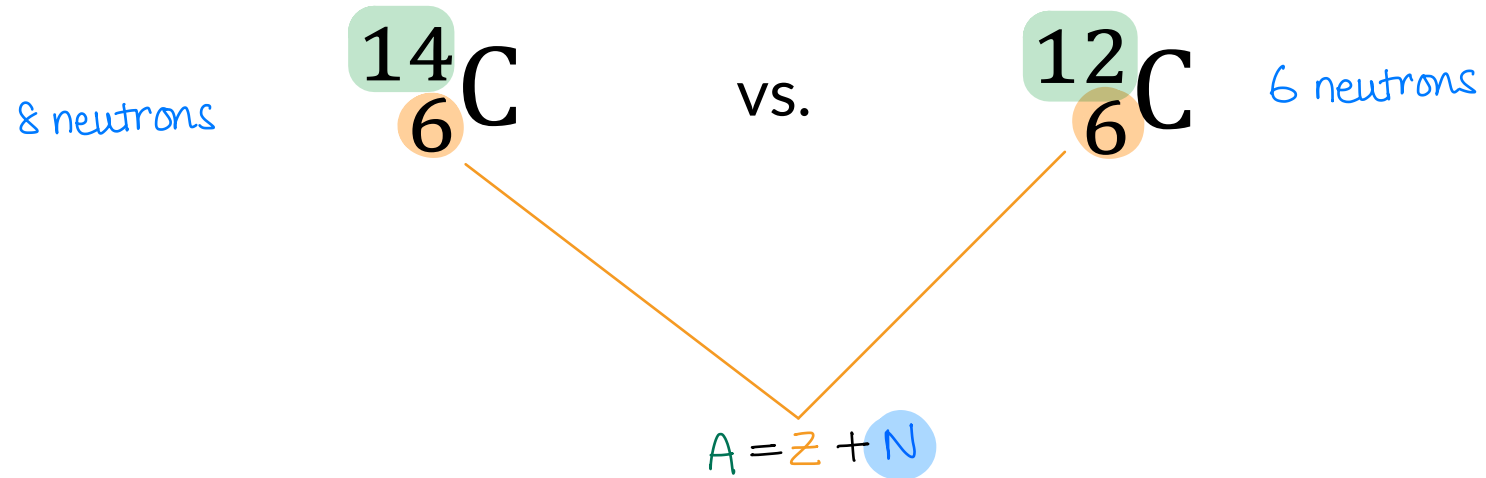
Mass number, A , is the number of protons plus the number of neutrons in an atom.

Subatomic Particles, Isotopes, and Ions

2.4

Isotopes are atoms of the ^{same Z} same element that have different numbers of neutrons.

↳ ergo, different A values



EXAMPLE PROBLEM 1

Learning Objective(s): 2.5

An atom of ^{210}Pb has how many neutrons?

$$\begin{array}{c} 210 \\ | \\ A = Z + N \end{array}$$

Handwritten: 82 (with a curved line pointing to Z)

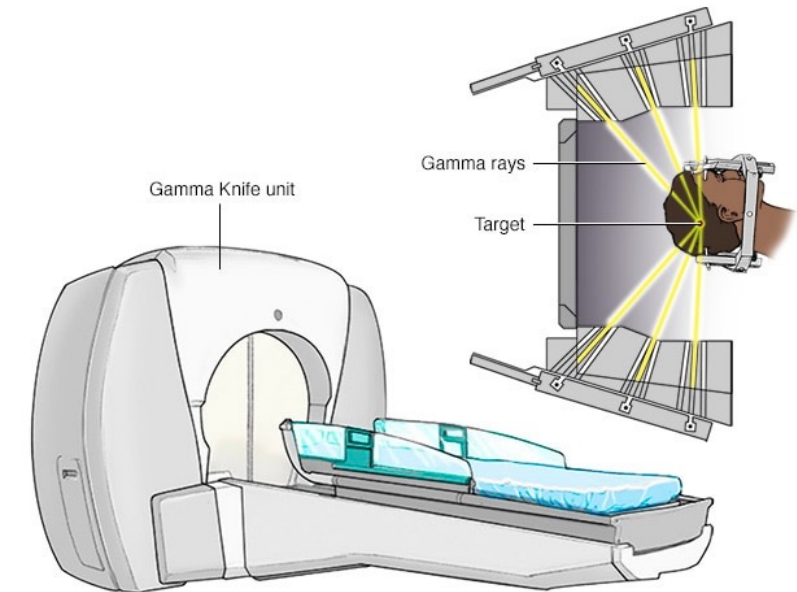
$$\# \text{ neutrons} = N = 210 - 82 = 128 \text{ neutrons}$$

How many neutrons are there in an atom of the isotope ^{60}Co ?

$$\begin{array}{c} 60 \\ | \\ A = Z + N \end{array}$$

Handwritten: 27 (with a curved line pointing to Z)

$$\# \text{ neutrons} = N = 60 - 27 = 33 \text{ neutrons}$$

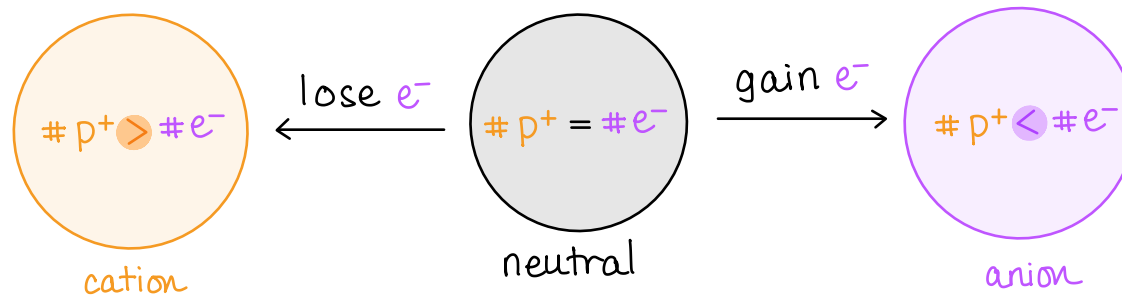


Subatomic Particles, Isotopes, and Ions

2.4

Ions: Species with an overall net charge because the numbers of protons and electrons are not equal.

- Anions: more electrons than protons net *negative* charge
- Cations: more protons than electrons net *positive* charge



Ions can be either monatomic (derived from a single atom) or polyatomic (a group of atoms with an overall charge).

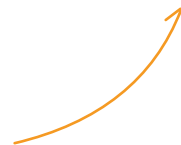
IN-CLASS QUESTION 1

Learning Objective(s): 2.5

$$A = Z + N$$

Determine the number of protons, neutrons, and electrons in the ion below.

from PT : O has $Z = 8$



neutral ^{15}O : 8 protons
7 neutrons
8 electrons

$^{15}\text{O}^{2-}$: 8 protons
7 neutrons
10 electrons



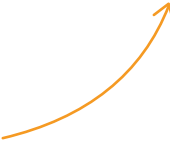
gain $2e^-$

EXAMPLE PROBLEM 2

Learning Objective(s): 2.1, 2.4

How many protons and electrons are in the chlorine atom? In the chlorine ion (Cl^-)?

from PT : Cl has $z = 17$



neutral
Cl

17 protons

17 electrons

anion
 Cl^-

17 protons

18 electrons

—————→
gain $1e^-$

Atomic Masses

2.5

The masses of atoms that we use today are on a relative scale.

- The mass of one isotope of one element was assigned a value, and the masses of all other atoms are measured relative to that standard.

Average atomic mass

- The weighted average of the masses of the naturally occurring mixture of isotopes of an element, compared with one-twelfth of the mass of a single ^{12}C atom.
- Found on the periodic table along with the atomic number for every element.
- *Also called: atomic mass, atomic weight, or average atomic weight.*

average atomic mass of element

mass of isotope

fractional abundance of isotope: $0 \leq f_i \leq 1$

$$\bar{m} = \sum_i m_i \times f_i$$
$$\sum_i f_i = 1$$

EXAMPLE PROBLEM 3

Learning Objective(s): 2.6

Antimony, Sb, has two stable isotopes with the following atomic masses (in amu):

$$\begin{array}{lcl} {}^{123}\text{Sb} = 122.904 \text{ amu} = m_1 & ; & f_1 = ? \\ {}^{121}\text{Sb} = 120.868 \text{ amu} = m_2 & ; & f_2 = ? \end{array} \quad \left. \vphantom{\begin{array}{l} f_1 = ? \\ f_2 = ? \end{array}} \right\} \begin{array}{l} f_1 + f_2 = 1 \\ f_2 = 1 - f_1 \end{array}$$

Determine the isotopic (fractional) abundance of ${}^{123}\text{Sb}$ using the average atomic mass of Sb from the periodic table.

from PT, $\bar{m}_{\text{Sb}} = 121.76 \text{ amu}$

$$\bar{m} = \sum_i m_i \times f_i$$

$$\bar{m} = m_1 f_1 + m_2 f_2 \quad ; \quad f_2 = 1 - f_1$$

$$121.76 \text{ amu} = (122.904 \text{ amu})(f_1) + (120.868 \text{ amu})(1 - f_1)$$

$$f_1 = 0.438$$

IN-CLASS QUESTION 2

Learning Objective(s): 2.6

There are three naturally occurring isotopes of silicon (atomic masses listed below). The percent abundance of ^{29}Si is 4.68 %.

Find the percent abundance of ^{30}Si using the following information and the average atomic mass of Si from the periodic table.

$$m_1 = {}^{28}\text{Si} = 27.9769 \text{ amu}, f_1 = ?$$

$$m_2 = {}^{29}\text{Si} = 28.9765 \text{ amu}, f_2 = 0.0468$$

$$m_3 = {}^{30}\text{Si} = 29.9738 \text{ amu}, f_3 = ?$$

$$f_1 + f_2 + f_3 = 1$$

$$f_1 = 1 - f_2 - f_3$$

$$f_1 = 0.9532 - f_3$$

$$\bar{m} = \sum_i m_i \times f_i$$

$$\bar{m} = m_1 f_1 + m_2 f_2 + m_3 f_3$$

$$\begin{aligned} 28.085 \text{ amu} &= (27.9769 \text{ amu})(0.9532 - f_3) \\ &\quad + (28.9765 \text{ amu})(0.0468) \\ &\quad + (29.9738 \text{ amu})(f_3) \end{aligned}$$

$$f_3 = 0.0307 \text{ or } 3.07 \%$$

The Periodic Table

2.6

The periodic table organizes elements by increasing atomic number with elements having similar chemical properties aligned in vertical columns.

Groups: The vertical columns of the periodic table

- The modern periodic table labels the groups 1-18.
- Older versions use 1A-2A for the main group elements and 1B-8B for the transition metals. Roman numerals also are used frequently in older versions.

Periods: The horizontal rows of the periodic table

ОПЫТЪ СИСТЕМЫ ЭЛЕМЕНТОВЪ.

ОСНОВАННОЙ НА ИХЪ АТОМНОМЪ ВѢСѢ И ХИМИЧЕСКОМЪ СХОДСТВѢ.

			Ti = 50	Zr = 90	? = 180.
			V = 51	Nb = 94	Ta = 182.
			Cr = 52	Mo = 96	W = 186.
			Mn = 55	Rh = 104,4	Pt = 197,1.
			Fe = 56	Ru = 104,4	Ir = 198.
			Ni = Co = 59	Pd = 106,6	Os = 199.
			Cu = 63,4	Ag = 108	Hg = 200.
H = 1	Be = 9,1	Mg = 24	Zn = 65,2	Cd = 112	
	B = 11	Al = 27,1	? = 68	Ur = 116	Au = 197?
	C = 12	Si = 28	? = 70	Sn = 118	
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?
	O = 16	S = 32	Se = 79,4	Te = 128?	
	F = 19	Cl = 35,5	Br = 80	I = 127	
Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204.
		Ca = 40	Sr = 87,6	Ba = 137	Pb = 207.
		? = 45	Ce = 92		
		?Er = 56	La = 94		
		?Yt = 60	Di = 95		
		?In = 75,6	Th = 118?		

Д. Менделѣевъ

FIGURE 2.15 Mendeleev's Periodic Table

PERIODIC TABLE OF ELEMENTS

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18				
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026				
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122															5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305															13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798				
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29				
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)				
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)				
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																						
				6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97			
				7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)			

Ptable



The Periodic Table

2.6

The elements also can be grouped in larger categories:

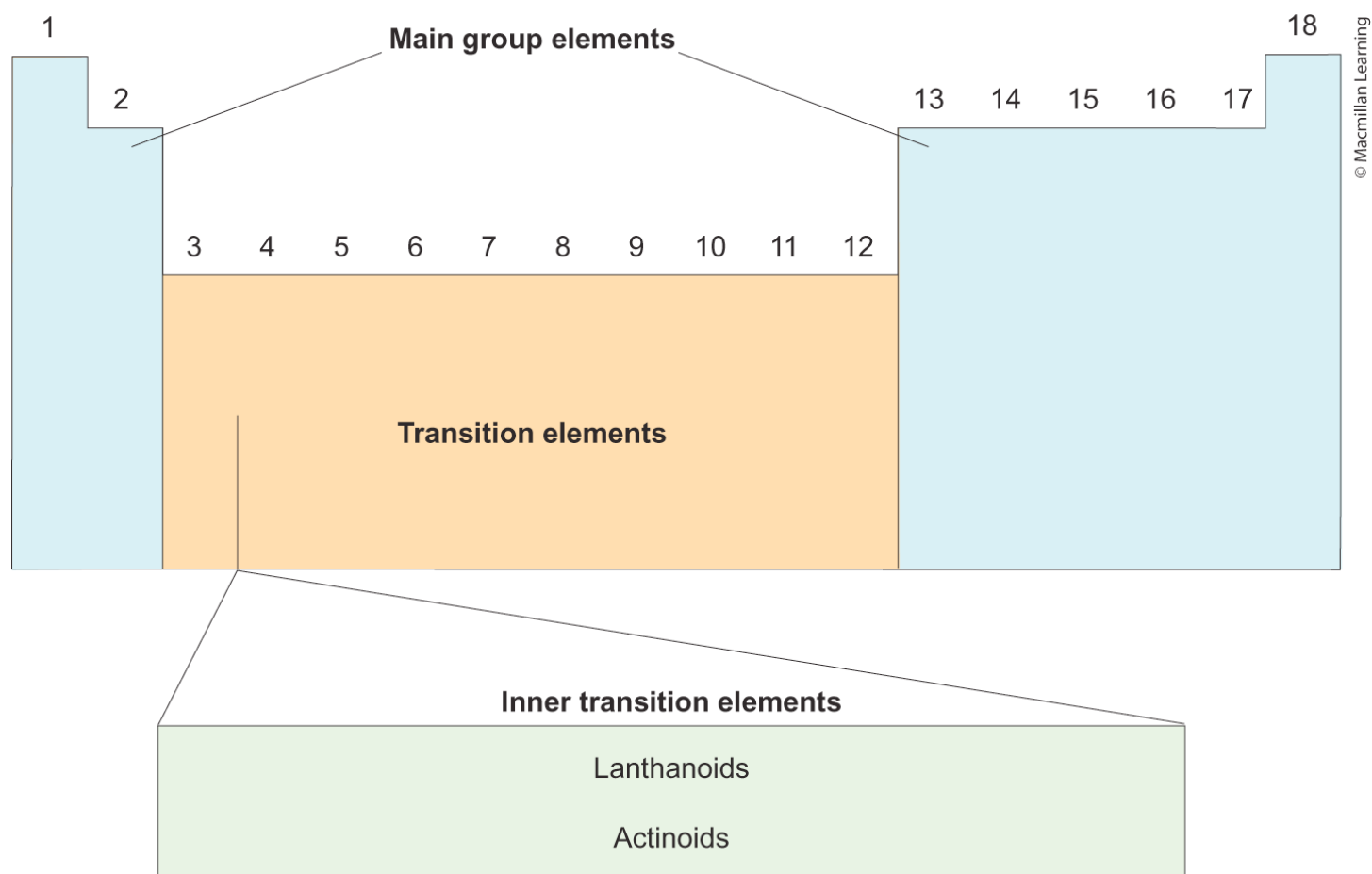


FIGURE 2.20

The Periodic Table

2.6

The elements also can be grouped in larger categories:

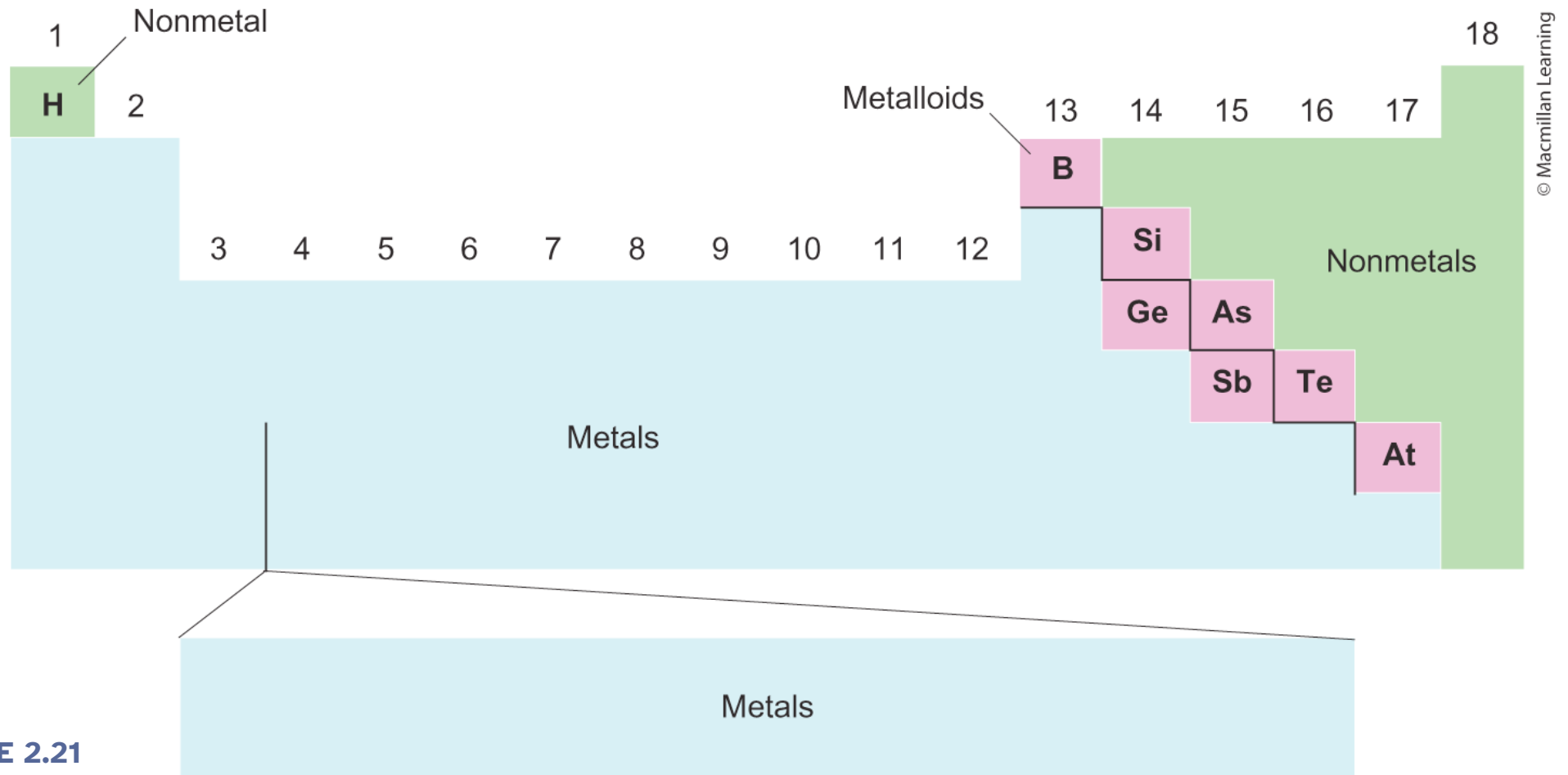


FIGURE 2.21

IN-CLASS QUESTION 3

Learning Objective(s): 2.7, 2.8

Which of the following are classified as nonmetals?

☐ Si

semimetal or metalloid

☒ Ne

nonmetal

☐ Fe

metal

☐ Sr

metal

☒ Br

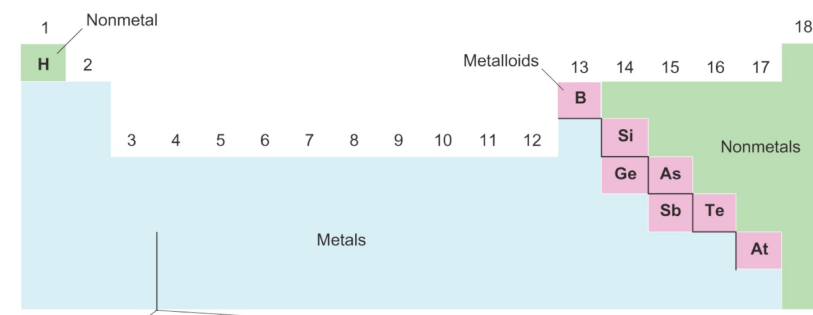
nonmetal



silicon nugget



Br₂ liquid



IN-CLASS QUESTION 4

Learning Objective(s): 2.7, 2.8

What is the chemical symbol of the alkali metal in the fourth period? potassium

column/group 1 = alkali metals

row/period 4 → 4

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.008																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminum 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 Lanthanum series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103 Actinide series	104 Rf Rutherfordium (261)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (260)

Ptable.com

Terminology

Chemistry in Context

It's important to understand chemical vocabulary.

Atom: Smallest part of an element that is still that element.

Molecule: Two or more atoms joined together and acting as a unit.

Compound: Two or more *different* elements joined together.

Chemists/Scientists use the term element in different ways though.

Chemical Formulas

3.1

Atomic elements: Most elements exist in nature as individual neutral atoms.

Molecular elements: A few elements exist in nature as small molecules containing two or more atoms bonded together.



(a)



(b)



(c)

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FIGURE 3.2 Examples of molecular elements (a) Cl_2 , (b) P_4 , (c) S_8 .

Chemical Formulas

3.1

Molecular diatomic elements: Seven elements occur as diatomic molecules when they are in their elemental/natural form.

- Hydrogen = H_2
- Nitrogen = N_2
- Oxygen = O_2
- Fluorine = F_2
- Chlorine = Cl_2
- Bromine = Br_2
- Iodine = I_2

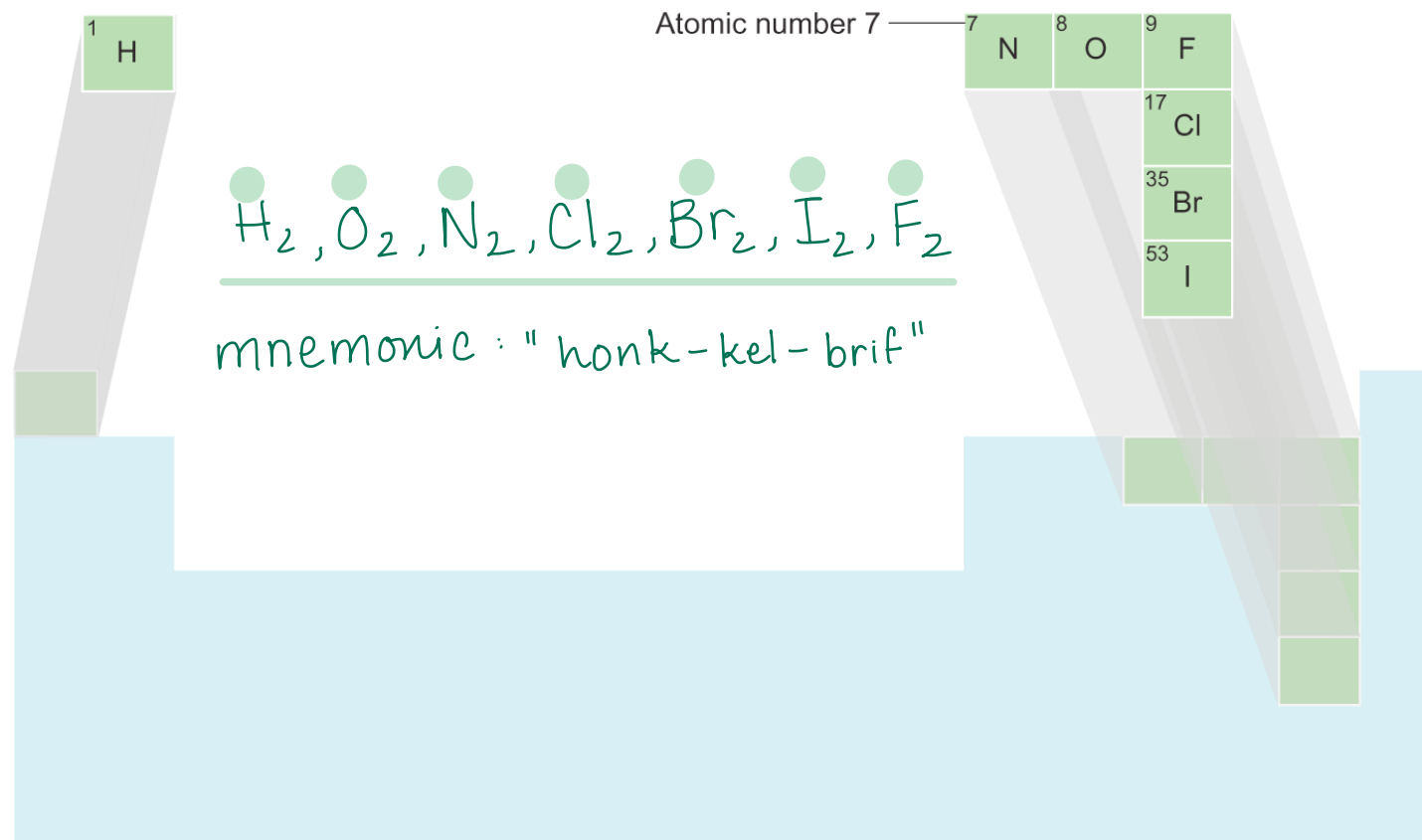


FIGURE 3.3 The Seven Elements that Exist as Diatomic Molecules

Chemical Formulas

3.1

Chemical formulas combine elemental symbols and subscripts to represent chemical compounds.

- Provide information about the relative number of atoms of each element
- Sometimes also give information about the arrangement of atoms in the molecule.

The group of atoms represented by a chemical formula is called a formula unit.

- An atom or molecule of an uncombined element, a molecule of a covalent compound, or a set of ions in an ionic compound

Chemical Formulas

3.1

Compounds can exist in different forms.

- Molecular or covalent compounds: Exist as individual particles (called molecules) made up of bonded nonmetal elements.
- Ionic compounds: Exist not as individual particles but as extended, 3D lattice structures. The simplest repeating unit is called a formula unit.

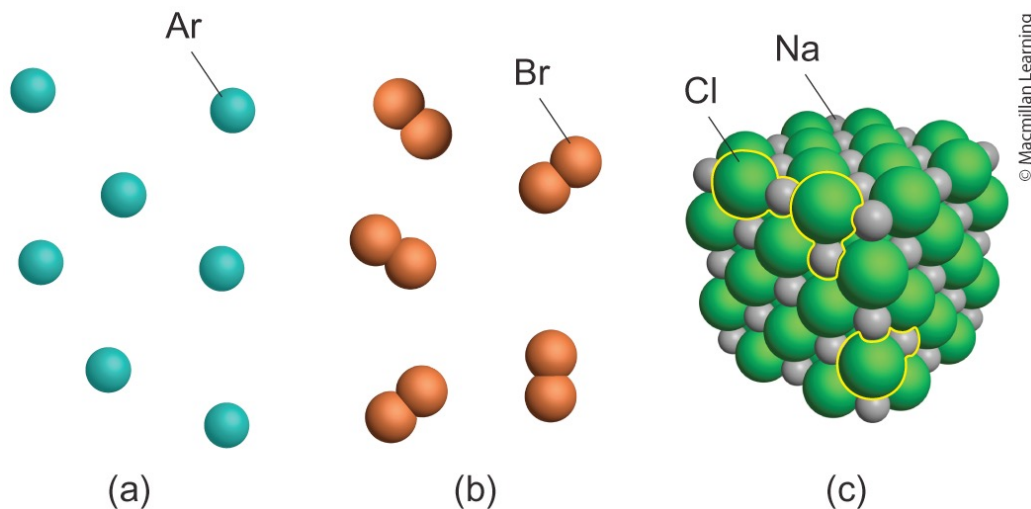


FIGURE 3.1 (a) atoms of Ar,
(b) molecules of Br_2 ,
(c) lattice of NaCl

Chemical Formulas

3.1

Chemists combine chemical symbols with subscripted numbers to form chemical formulas.

The subscripted numbers indicate the number of atoms of each element present.

